ECON 770 Recitation 2

Aug 21, 2020

University of North Carolina at Chapel Hill

Reminder

- Due of Assignment #1: Aug 25, 12:00 PM
- "Turn in your code by emailing[1] the TA the program: give the email a subject line **770 Assign. 1 code**. This program must work when the TA runs it in the software of your choice." (I mainly use Python and STATA, but still I can read Matlab, R and Julia)

[1] Hyperlink: tianqili@ad.unc.edu

Contents

- Conditional Probability
- Bayes' Rule
- Arrangement, Permutation and Combination
- Random Variables, CDF, Discrete PMF

Conditional Probability

Properties

- $P(A|B) \geq 0$
- $P(B|B) = 1$ (as $P(B|B) = \frac{P(B \cap B)}{P(B)} = \frac{P(B)}{P(B)}$)
- If A_i are disjoint events, then $P\left(\cup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P\left(A_i \mid B\right)$
- $P(A|B) \neq P(B|A)$ if $P(A) \neq P(B)$ or $P(A \cap B) = 0$
- $P(A) = P(A \cap \Omega) = P\left(A \cap (B \cup B^c)\right) = P(A \cap B) + P(A \cap B^c)$
- $P(A^c \mid B) = 1 - P(A \mid B)$

How to show $P(A|B) \neq P(B|A)$ if $P(A) \neq P(B)$?

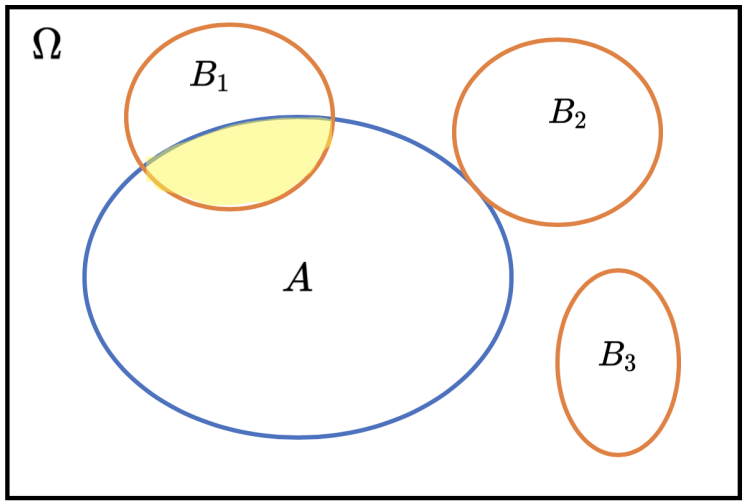
Proof: By definition,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ versus } \frac{P(B \cap A)}{P(A)} = P(B|A)$$

where $P(A \cap B) = P(B \cap A)$, however, ...

Conditional Probability (Typo)

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i) \text{ for disjoint } \{B_i\}$$

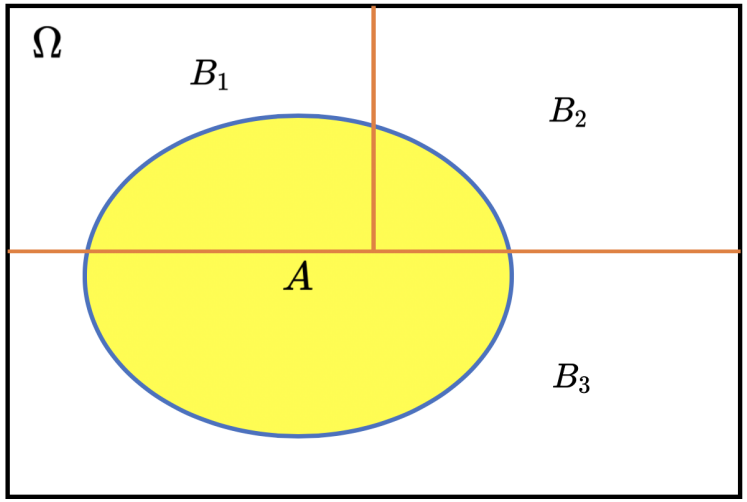


but

$$P(A) \neq \sum_{i=1}^{n} P(A \mid B_i) P(B_i) = \sum_{i=1}^{n} \frac{P(A \cap B_i)}{P(B_i)} P(B_i) = \sum_{i=1}^{n} P(A \cap B_i)$$

Conditional Probability (Revised)

$$P(A) = \sum_{i \in I} P(A \mid B_i) P(B_i) \text{ for partition } \bigcup_{i \in I} B_i = \Omega$$



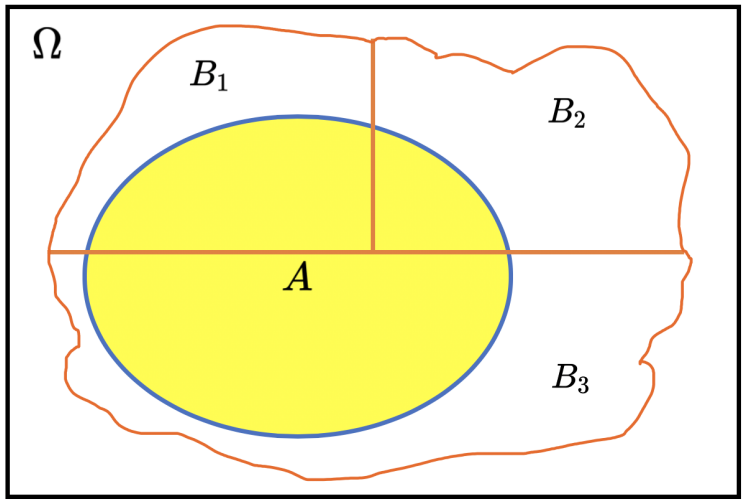
because it directly comes from

$$P(A) = \sum_{i \in I} P(A \cap B_i) \text{ where } B_i \cap B_j = \varnothing \text{ if } i \neq j \text{ and } \bigcup_{i \in I} B_i = \Omega$$

Conditional Probability (Relaxed)

For any $A \subseteq B \subseteq \Omega$,

$$P(A) = \sum_{i \in I} P(A \mid B_i) P(B_i) \text{ for partition } \bigcup_{i \in I} B_i = B$$



Bayes' Rule

Theorem

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)} \text{ for any partition } \{A_i\} \text{ of } \Omega$$

Proof.

By definition of conditional probability,

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i)P(A_i)}{P(B)}$$

Since we have proved that for any partition $\{A_i\}$ s.t. $\bigcup_i A_i = \Omega$

$$P(B) = \sum_i P(B|A_i)P(A_i)$$

then we plug it in denominator and ... □

Proof of Binomial Theorem*

Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Proof.

We prove this by mathematical induction. When $n = 1$, this is quite obvious. Suppose that this formula is true for $n = m \geq 1$, then, when $n = m + 1$, we have

$$\begin{aligned}
(a+b)^{m+1} &= a(a+b)^m + b(a+b)^m \\
&= a \textstyle\sum_{k=0}^{m} \binom{m}{k} a^{m-k} b^k + b \sum_{j=0}^{m} \binom{m}{j} a^{m-j} b^j \\
&= \textstyle\sum_{k=0}^{m} \binom{m}{k} a^{m-k+1} b^k + \sum_{j=0}^{m} \binom{m}{j} a^{m-j} b^{j+1}
\end{aligned}$$

Proof of Binomial Theorem*

Proof.

$$= a^{m+1} + \sum_{k=1}^{m} \binom{m}{k} a^{m-k+1} b^k + \sum_{j=0}^{m} \binom{m}{j} a^{m-j} b^{j+1}$$

$$= a^{m+1} + \sum_{k=1}^{m} \binom{m}{k} a^{m-k+1} b^k + \sum_{k=1}^{m+1} \binom{m}{k-1} a^{m-k+1} b^k$$

$$= a^{m+1} + \sum_{k=1}^{m} \binom{m}{k} a^{m-k+1} b^k + \sum_{k=1}^{m} \binom{m}{k-1} a^{m+1-k} b^k + b^{m+1}$$

$$= a^{m+1} + b^{m+1} + \sum_{k=1}^{m} \left[\binom{m}{k} + \binom{m}{k-1} \right] a^{m+1-k} b^k$$

$$= a^{m+1} + b^{m+1} + \sum_{k=1}^{m} \binom{m+1}{k} a^{m+1-k} b^k$$

$$= \sum_{k=0}^{m+1} \binom{m+1}{k} a^{m+1-k} b^k \qquad \text{as } \binom{n-1}{k} + \binom{n-1}{k-1} = \binom{n}{k}$$

Combinatorics

Theorem

(Possibility and Probability) For an event $A \in \sigma(\Omega)$ where Ω s.t. $P(\omega_i) = P(\omega_j)\ \forall\ i, j,\ \omega_i \in \Omega$, then

$$P(A) = \frac{\#(A)}{\#(\Omega)}$$

where $\#(A)$ is the number of outcomes in A

Example: Roll n fair dice, flip a fair coin n times, ...

Arrangement, Permutation, and Combination

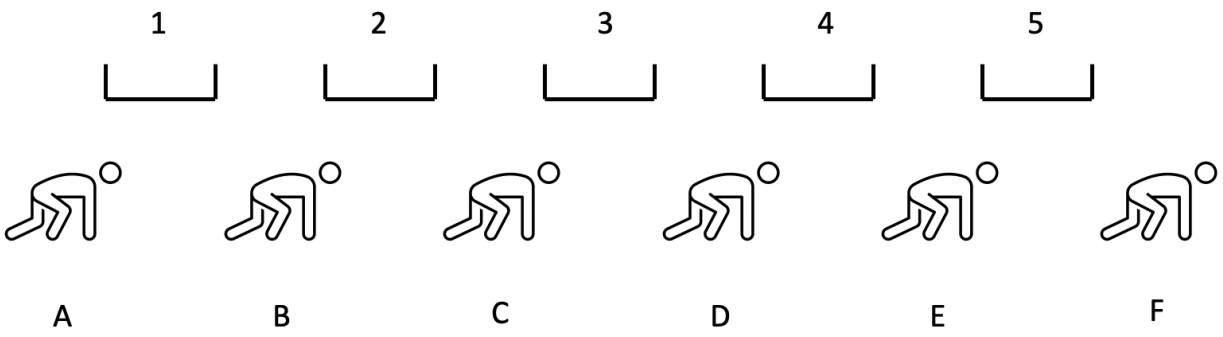
Definition and Proposition

- $\#(arrangement) = n!$ (not necessary unique)
- $\#(permutations) \equiv P_r^n = \dfrac{n!}{(n-r)!}$ (distinct and ordered)
- $\#(combination) \equiv C_r^n = \dfrac{n!}{(n-r)!r!}$ (distinct but not ordered)

* Typos in lecture slides are highlighted and amended in red

Example

Imagine that we have five empty seats here, and there are six students



- All possible arrangements: $6 \times 5 \times 4 \times 3 \times 2$

Example

Suppose that we have $\{a, b, c, d\}$, then

- Pick up 1 from 4, ordered: $P_1^4 = \frac{4!}{3!} = 4$

$$\{a\}, \{b\}, \{c\}, \{d\}$$

- Pick up 2 from 4, ordered: $P_2^4 = \frac{4!}{2!} = 12$

$$\{a,b\}, \{a,c\}, \{a,d\}, \{b,a\}, \{b,c\}, \{b,d\},$$
$$\{c,a\}, \{c,b\}, \{c,d\}, \{d,a\}, \{d,b\}, \{d,c\}.$$

- Pick up 3 from 4, ordered: $P_3^4 = \frac{4!}{1!} = 24$

$$\{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{a,c,b\}, \{a,d,b\}, \{a,d,c\},$$
$$\{b,a,c\}, \{b,a,d\}, \{b,c,d\}, \{b,c,a\}, \{b,d,a\}, \{b,d,c\},$$
$$\{c,a,b\}, \{c,a,d\}, \{c,b,d\}, \{c,b,a\}, \{c,d,a\}, \{c,d,b\},$$
$$\{d,a,b\}, \{d,a,c\}, \{d,b,c\}, \{d,b,a\}, \{d,c,a\}, \{d,c,b\}.$$

Example

Suppose that we have $\{a, b, c, d\}$, then

- Pick up 1 from 4, not ordered: $C_1^4 = \dfrac{4!}{3!} = 4$

$$\{a\}, \{b\}, \{c\}, \{d\}$$

- Pick up 2 from 4, not ordered: $C_2^4 = \dfrac{4!}{2!2!} = 6$

$$\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\},$$

$$\cancel{\{b, a\}, \{c, a\}, \{d, a\}, \{c, b\}, \{d, b\}, \{d, c\}}.$$

- Pick up 3 from 4, not ordered: $C_3^4 = \dfrac{4!}{1!3!} = 4 = C_1^4$

$$\{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}$$

Note that $\begin{pmatrix} m \\ k \end{pmatrix}$ can be written as C_k^n, ${}_nC_k$, $C_{n,k}$ or even C_n^k

Random Variables

Defn.

A random variable X is a mapping from the sample space to the reals, i.e.

$$X : \Omega \to \mathbb{R}$$
$$\omega \mapsto a$$

Example:

- Suppose there's die $\Omega = \{1, 2, \ldots, 6\}$, define $X(\omega) = -\omega$, therefore

$$X(1) = -1,\ X(2) = -2, \ldots,\ X(6) = -6$$

- Suppose that $\Omega = [0, +\infty)$ is the space of salary, then we can define a binary variable X for each income $\omega \in \Omega$ as

$$X(\omega) = \mathbb{1}\{\omega \geq 10000\}$$

Cumulative Distribution Function (CDF)

Defn.

A cumulative distribution function $F : \mathbb{R} \to [0, 1]$ of random variable X is a function s.t. $\forall x \in \mathbb{R}$

$$F(x) = P(X(\omega) \leq x)$$

In 2-dim case, $F(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2)$

Example

$$F_X(x) = \begin{cases} 0 & : x < 0 \\ x & : 0 \leq x \leq 1 \\ 1 & : x > 1 \end{cases} \qquad F_X(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$F(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) dt$$

CDF Examples

Example: Roll two fair dice $\omega = (a, b) \in \Omega = \{1, \ldots, 6\} \times \{1, \ldots, 6\}$ and define

$$X(\omega) = \mathbb{1}\{a = b\}$$

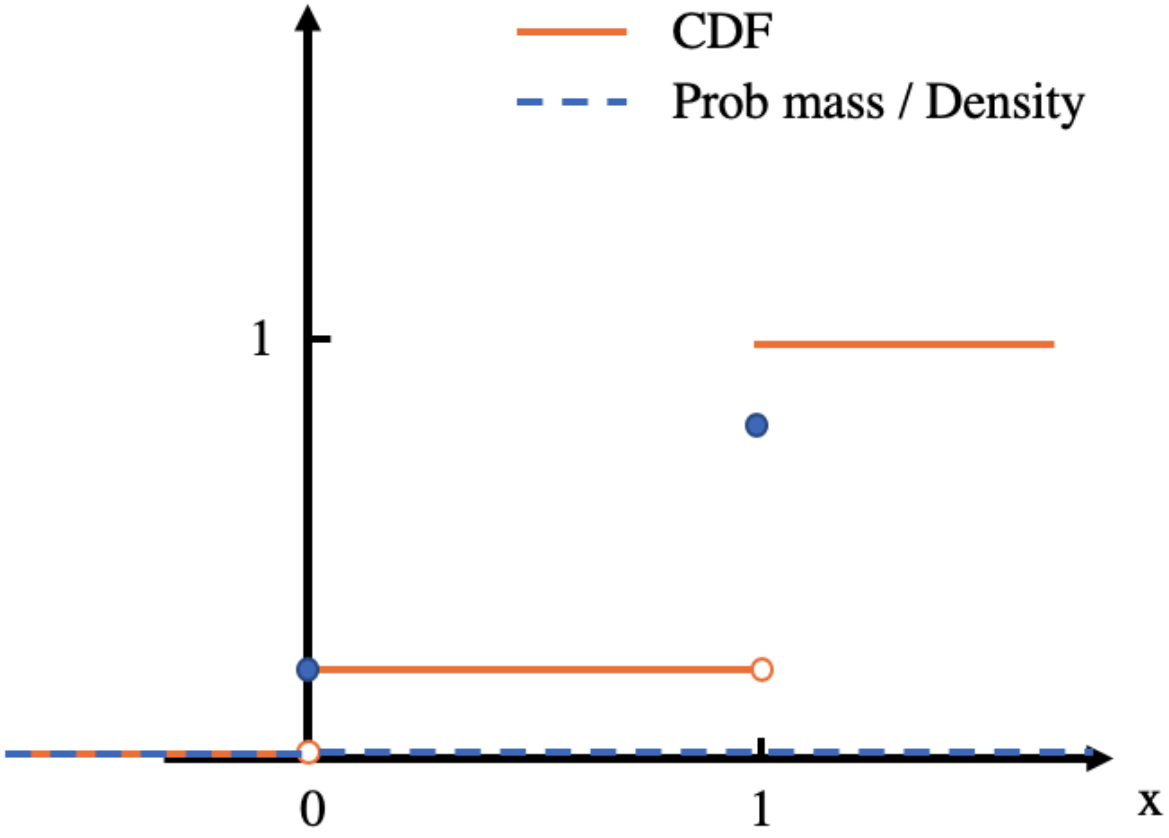
		Die 2					
		1	2	3	4	5	6
Die 1	1	1/36	1/36	1/36	1/36	1/36	1/36
	2	1/36	1/36	1/36	1/36	1/36	1/36
	3	1/36	1/36	1/36	1/36	1/36	1/36
	4	1/36	1/36	1/36	1/36	1/36	1/36
	5	1/36	1/36	1/36	1/36	1/36	1/36
	6	1/36	1/36	1/36	1/36	1/36	1/36

Then, for any $x < 0$, we have $F(x) = 0$ obviously.

$F(0) = P(X(\omega) \leq 0) = P(\{X < 0\} \cup \{X = 0\}) = P(a \neq b) = \frac{30}{36} = \frac{5}{6}$

$F(1) = P(X(\omega) \leq 1) = P(\{X \leq 0\} \cup \{0 < X < 1\} \cup \{X = 1\}) = \frac{5}{6} + \frac{1}{6} = 1$

CDF Examples



CDF Properties

Theorem

- $F(x) = P(X \leq x) \in [0, 1]$
- $\lim_{x \to \infty} F(x) = 1$
- $\lim_{x \to -\infty} F(x) = 0$
- *Non-decreasing:* $\forall\ x \leq y,\ F(x) \leq F(y)$
- *Right continuous:* $\forall\ x,\ \lim_{\delta \searrow 0} F(x + \delta) = F(x)$

Exercise 1

(Casella and Berger, 1.49) A cdf F_X is *stochastically greater* than a cdf F_Y if $F_X(t) \leq F_Y(t)$ for all t and $F_X(t) < F_Y(t)$ for some t. Prove that if $X \sim F_X$ and $Y \sim F_Y$, then

$$P(X > t) \geq P(Y > t) \text{ for every } t$$

and

$$P(X > t) > P(Y > t) \text{ for some } t$$

that is, X tends to be bigger than Y

Random Variables (Discrete)

Definition

$X(\omega)$ is discrete if it has countably[a] many possible values. Suppose that $F(X)$ is a CDF defined on $X(\omega)$, then

$$F(x) = P(X \leq x) = \sum_{a \leq x,\ a \in S} P(X = a)$$

is a Càdlàg function, where $P(X = a)$ is called probability mass function (PMF) and S is called support, the set of all possible values of x s.t. $P(X = x) > 0$

[a] finite or countably infinite

Theorem

- $\forall\, x \in \mathbb{R}\,,\ 0 \leq P(X = x) \leq 1$
- $\sum_{x \in S} P(X = x) = 1$